

# Candidate Data Transformer

## Multi-Source Candidate Profile Normalization Pipeline

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### Project Objective

This system normalizes fragmented candidate details from Resume PDFs and Recruiter CSVs into a single canonical profile. The pipeline performs parsing, normalization, identity consistency validation, merge engine resolution, confidence scoring, provenance tracking, and schema validation. This deterministic architecture ensures data consistency, explainability, and downstream search usability.

### System Pipeline



#### Key Features

- Deterministic candidate pipeline.
- Resume PDF parsing using pdfplumber.
- Recruiter CSV ingestion and field mapping.
- Canonical field normalization.
- Identity consistency validation.
- Configurable override merge logic.
- Confidence scoring based on source agreement.
- Explainable outputs with field-level provenance.
- Config-driven JSON projections.
- Draft-07 JSON Schema validation.

#### Technology Stack

- Python
- Streamlit
- Pandas
- pdfplumber
- RapidFuzz
- Pydantic
- JSON
- YAML

#### Outputs

- Canonical Candidate Profile JSON
- Configured Projected JSON Output
- Descriptive Stage Ingestion Summary
- Detailed Confidence Scores Matrix
- Field-Level Provenance Trace Logs
- JSON Schema Validation Reports

#### Results

- Successfully transforms Resume-only candidate profiles.
- Successfully transforms Recruiter CSV candidate profiles.
- Supports deterministic multi-source profile merging.
- Gracefully handles incomplete candidate information.
- Prevents unsafe merges after identity validation.
- Produces explainable outputs with complete provenance tracking.